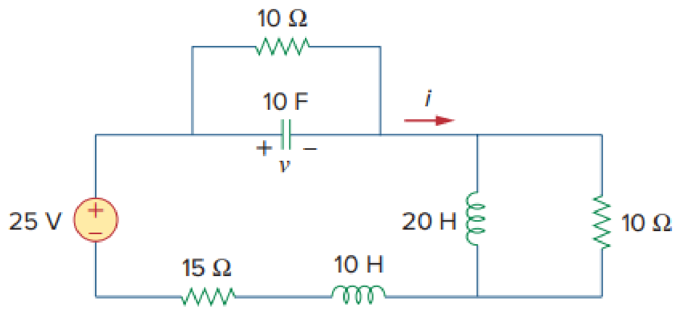
Chapter 6, Problem 48P

Problem

Under steady-state dc conditions, find *i* and *v* in the circuit in Fig. 6.71.

Fig. 6.71:



Step-by-step solution

Step 1 of 3

Under steady state dc condition Inductor acts like short circuit and capacitor acts like open circuit. So the circuit can be modified as

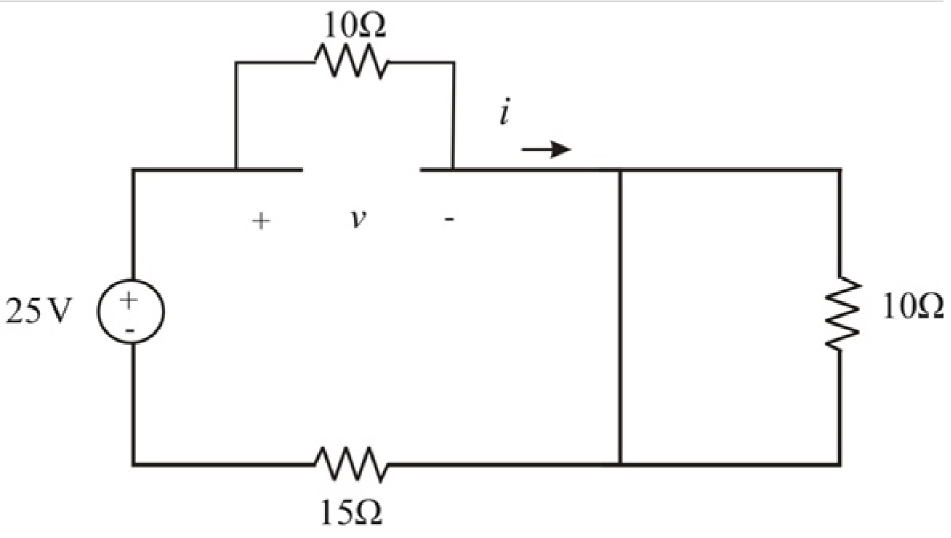


Figure 1

Step 2 of 3

From Figure (1) short circuit is in parallel with 10Ω resistor so remove the 10Ω resistor.

Redraw the circuit:

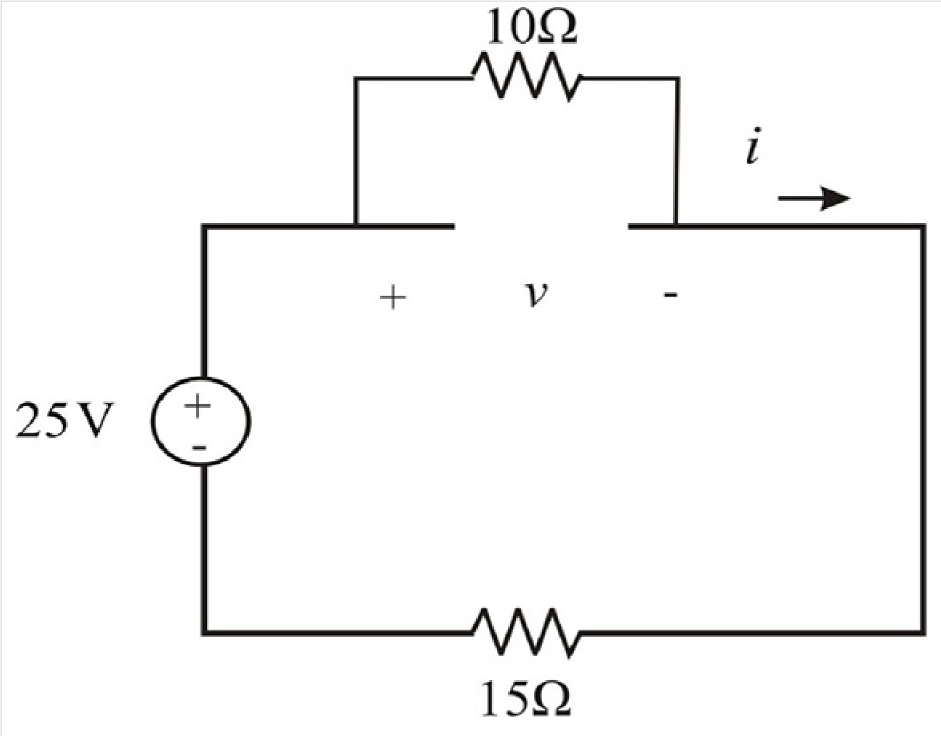


Figure 2

Step 3 of 3

Determine the equivalent resistance of the circuit.

$$\begin{aligned} R_{eq} &= 10\,\Omega + 15\,\Omega \\ &= 25\,\Omega \end{aligned}$$

Calculate the current through the circuit.

$$i = \frac{v}{R_{eq}}$$

$$i = \frac{25}{25}$$

$$i = 1\text{ A}$$

Calculate the voltage across the capacitor.

Since, $10\,\mu\text{F}$ capacitor is parallel with $10\,\Omega$ resistor. The voltage across the capacitor is same as the voltage across the resistor.

Therefore, the voltage across $10\,\Omega$ resistor is,

$$v = 10i$$

$$v = 10 \times 1$$

$$v = 10\text{ V}$$

Hence, the voltage and current under steady state dc condition for the given circuit is

$$\boxed{\begin{array}{l} v = 10\text{ V} \\ i = 1\text{ A} \end{array}}.$$